

Deep learning with reflection high-energy electron diffraction images to predict cation ratio in $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ thin films

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ABSTRACT

Machine learning (ML) with *in situ* diagnostics offers a transformative approach to accelerate, understand, and control thin film synthesis by uncovering relationships between synthesis conditions and material properties. In this study, we demonstrate the application of deep learning to predict the stoichiometry of $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ thin films using reflection high-energy electron diffraction images acquired during pulsed laser deposition. A gated convolutional neural network trained for regression of the Sr atomic fraction achieved accurate predictions with a small dataset of 31 samples. Explainable AI techniques revealed a previously unknown correlation between diffraction streak features and cation stoichiometry in $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ thin films. Our results demonstrate how ML can be used to transform a ubiquitous *in situ* diagnostic tool, that is usually limited to qualitative assessments, into a quantitative surrogate measurement of continuously valued thin film properties. Such methods are critically needed to enable real-time control, autonomous workflows, and accelerate traditional synthesis approaches.

KEYWORDS: pulsed laser deposition, deep learning, RHEED, *in situ* diagnostics

In recent years, machine learning (ML) has emerged as a promising tool in materials synthesis to discover or optimize the synthesis procedure for targeted structure, composition, or properties¹⁻³. For synthesis, ML is typically used in an active-learning framework⁴⁻⁷ to efficiently explore large synthesis parameter spaces and achieve the desired result in fewer experiments than traditional methods. In this approach, synthesis recipes are used to train ML models to predict one or more desired film characteristics, and Bayesian optimization is commonly used for sequential design of experiments. ML models can capture complex relationships in the data that correlate synthesis conditions and/or diagnostic measurements to material characteristics that may be otherwise difficult to discover or require significant domain expertise to reveal.

Synthesis of thin film materials using physical vapor deposition (PVD) methods such as molecular beam epitaxy⁸ (MBE) and pulsed laser deposition⁹ (PLD) are a cornerstone of discovery in materials science, allowing for the synthesis and study of a wide variety of solid-state materials and interfaces. These techniques possess large parameter spaces and are traditionally time-consuming and labor intensive, making their workflows prime candidates for modernization through application of ML and automation. Fully autonomous synthesis with magnetron sputtering¹⁰ and PLD¹¹ have been realized by combining automated synthesis, characterization, and sample exchange equipment with ML but, so far, most PVD experiments are focused on using ML to streamline complex data analysis or assist in real-time control during growth of single samples using *in situ* diagnostics.

Historically, reflection high-energy electron diffraction¹² (RHEED) is a routine *in situ* diagnostic that can be used to examine the starting substrate and monitor the crystalline phase, orientation, and growth rate of subsequent thin films during MBE or PLD. In RHEED, a high energy (10-20 keV) beam of electrons is directed at the growing film's surface, where they interact

with the top 2-5 nm of the material and scatter to form a diffraction pattern on a phosphor screen, which is typically recorded with a camera. Oscillations in the intensity or width of the specular reflection or diffraction spots are commonly used to monitor the growth rate during layer-by-layer growth¹³. The full RHEED pattern contains a wealth of additional information but is often limited to qualitative assessments due to the challenges of accurately simulating experimental patterns and the absence of comprehensive databases linking RHEED patterns to film properties beyond simple crystal structure or surface reconstructions. Such observations have been used to estimate SrTiO₃ stoichiometry in the past via MBE and PLD growth¹⁴⁻¹⁶, but with a primary focus on whether films are Ti or Sr rich without quantifying the stoichiometry.

Most modern ML efforts with RHEED data have been for dimensionality reduction through matrix factorization techniques or pattern clustering¹⁷⁻²¹, classification of the surface structure²²⁻²⁵, semantic segmentation for diffraction pattern identification²⁶, and real-time changepoint identification²⁷. Regression methods for linking RHEED images to continuous quantities - e.g. stoichiometry or conductivity - with deep learning remains a little explored area but is promising in terms of transforming RHEED into a makeshift surrogate measurement for more time-consuming or complicated physical property measurements. If proven effective, such models can be used in autonomous synthesis workflows or for real-time adjustment of deposition conditions. Notably, multivariate linear models using manually engineered RHEED features (streak widths, centroid positions, area, etc.) have been found to be effective predictors of stoichiometry in W_{1-x}V_xSe₂ films with a small amount of data²⁸. Deep learning can circumvent the manual feature engineering step, which may fail to capture more subtle image features and relax the requirement for highly robust feature extraction (i.e. curve fitting), although its effectiveness has not yet been proven.

In this study, we seek to evaluate the efficacy of supervised deep learning with RHEED images for regression tasks. We trained a gated convolutional neural network (CNN) to predict the atomic fraction of Sr, x , from RHEED images of $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ thin films grown by PLD. We find that the model performs well on the regression task, predicting x with reasonable accuracy with a dataset of only 31 samples. Further, we use explainable AI techniques with the trained model to aid us in revealing a previously unknown correlation between the intensity ratio and spacing of the (01) and (02) diffraction peaks with the cation stoichiometry in SrTiO_3 . Thus, deep learning with RHEED images for regression tasks holds promise for prediction of other continuously valued film properties, rather than simple classifications, as well as an analysis tool to assist in interpreting RHEED pattern correlations to film properties. We anticipate that methods like this will become widely adopted as automated PLD and MBE equipment becomes more broadly available, which can produce a higher volume of reproducible samples for analysis that can reveal deeper insights to the synthesis process.

To evaluate the use of deep learning to predict the stoichiometry of thin films using RHEED images acquired during synthesis, we must generate our own data because no experimental database of well characterized materials with associated *in situ* diagnostics exists. We chose homoepitaxial SrTiO_3 as the test material system since its growth with PLD is well known and the cation stoichiometry is easily modulated via deposition from binary oxide targets. We synthesized 31 $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ films with various Sr/Ti ratios by sequentially depositing sub-monolayer thicknesses of SrO and TiO_2 from the respective targets. The procedure used here is similar to the one used by Herklotz et al.²⁹ RHEED was collected during deposition and the final images (after deposition was complete, at the deposition temperature) were used as the inputs to

the machine learning model. After deposition, the atomic fraction of Sr was measured by XPS and was used as the model's prediction target.

Figure 1a shows the experimental arrangement where the SrO and TiO₂ targets are moved into the path of the ablation laser by rotating the target carousel. First, we determined the deposition rate of SrO and TiO₂ under the same deposition conditions to be 0.790 Å/pulse and 0.1865 Å/pulse corresponding to 3 and 11 pulses per half-unit cell of SrTiO₃, respectively. The deposition sequence for each film was multiple cycles of 3 SrO pulses and 11+N pulses of TiO₂ where N ranged from -5 to 10 to control the relative Sr flux fraction provided to the substrate per cycle. Figure 1b schematically shows the laser pulse sequence on each target for a single deposition cycle. Two films with different nominal thicknesses of 15 nm and 20 nm were deposited for each N value to check the repeatability of the measured stoichiometry and RHEED patterns.

Figure 1c is a typical example of the RHEED specular reflection intensity versus time along the STO [110] direction, taken from the N = -1 sample grown to 20 nm thick using 47 deposition cycles. Generally, the specular intensity drops immediately upon the first SrO pulse and, after the first few deposition cycles, either increases or decreases during deposition from each target with some evolution in the time during target exchange where no flux is incident on the substrate. The inset of Figure 1c highlights an intensity increase during SrO deposition (green shaded areas) followed by a further increase during the target exchange time which saturates to some value. The intensity decreases during TiO₂ deposition (blue shaded areas) with, again, some evolution in intensity during the target exchange time. There is a lower frequency beat with a period of ~6-8 cycles (in this example) that appears due to the SrO deposition either causing the intensity to decrease and then increase (green shaded area at ~79 s) or only increase (green shaded area at ~87 s) corresponding to SrO deposition onto different layer coverages^{14,29} (i.e. incomplete half unit cell

layers). Since each SrO/TiO₂ deposition grows ~ 0.5 MLs of SrTiO₃, we attribute the changes in intensity during deposition to the typical RHEED oscillations observed with a layer-by-layer growth mode where the minima and maxima correspond to ~ 0.5 MLs and a completed MLs respectively.

RHEED patterns along the [110] azimuth of SrTiO₃ for the bare substrate, Sr rich ($N = -5$), stoichiometric ($N = -1$), and Ti rich ($N=7$) films are shown in Figure 1d-g. Past work has indicated that [110] azimuth is best for determination of stoichiometry in STO during shuttered MBE growth¹⁴. The films show a streaky pattern indicating a predominantly 2D surface and faint Kikuchi lines in Figure 1e-f suggest good crystal quality in some films. The arrows highlight the streaks from the (01) and (02) diffraction planes. Finally, we compare the Sr flux fraction to the film stoichiometry, atomic fraction Sr, measured by XPS. The Sr flux fraction is given by the expression $N_{Sr} / (N_{Sr} + N_{Ti})$ where N_{Sr} and N_{Ti} are the number of atoms deposited per cycle determined from the deposition rates and number of pulses on each target. Sr atomic fraction is given by the same expression but using the peak areas of the Sr 3d and Ti 2p XPS core-levels. The Sr flux fraction is directly proportional to the atomic Sr fraction, with a slope of 1.01 ± 0.031 (including trivial endpoints of 0.0 and 1.0). Thus, we have a simple control parameter to vary the

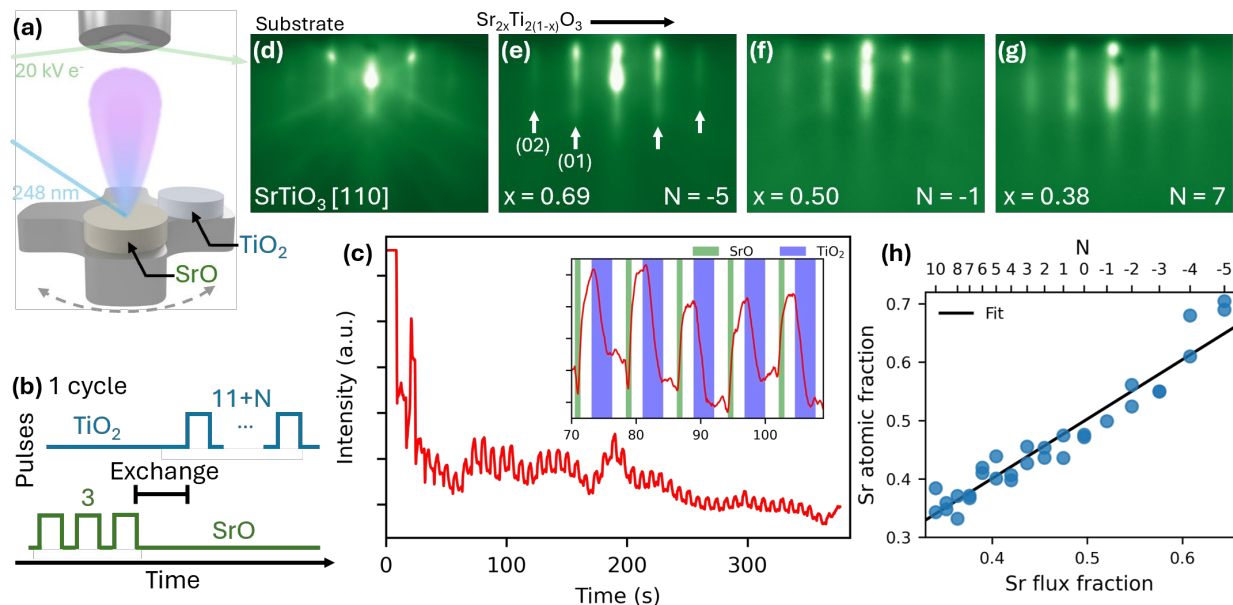


Figure 1. Deposition of $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ films by sequential pulsed laser deposition (PLD) of SrO and TiO_2 with reflection high-energy electron diffraction (RHEED) during growth. a) $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ films are grown by alternating sub-monolayer depositions from SrO and TiO_2 targets by exchanging target positions between deposition cycles. b) Repeated cycles of 3 pulses of SrO and $11+N$ pulses of TiO_2 are used to grow films to nominal thicknesses of 15 nm and 20 nm where N ranges from -5 to 10. c) RHEED specular reflection intensity vs time for $N = -1$ shows oscillations modulated by the PLD target exchange time. The inset details the intensity modulation during deposition from each target. d) RHEED along the $[110]$ azimuth of the SrTiO_3 substrate compared to the final RHEED image for e) Sr rich, f) stoichiometric, and g) Ti rich films along the same direction. The arrows indicate the (01) and (02) diffraction planes. h) The film Sr atomic fraction is directly proportional to the Sr flux fraction per deposition cycle with a fit slope of 1.01 ± 0.031 .

film stoichiometry, in this case, by selecting an N value to directly vary the amount of Ti supplied during growth of the TiO_2 sub-layer of SrTiO_3 .

Due to this direct correlation between the Sr flux and atomic fractions, the Sr flux fraction could be used as a “simple surrogate” machine learning target instead of the actual measured stoichiometry. This may be extensible to other materials synthesized through a similar binary target deposition method as an approximation for stoichiometry. However, for the most common single-target deposition scenario, it is not applicable. For example, the Sr/Ti ratio in films deposited from a pure STO target is known to change with the laser fluence and stoichiometry measurements would be required³⁰. The stoichiometry resulting from various N values was highly

repeatable, which is somewhat surprising considering the reproducibility problems that are known to exist in PLD and thin film growth in general. We ascribe the reproducibility, in this case, to our automated PLD system which carried out each deposition in a well-controlled, identical manner (at least in terms of the variables which were under our control). Additionally, the laser energy was measured *in vacuo* inside the chamber and adjusted to maintain a fluence of 1.0 J/cm^2 immediately before each deposition to remove the effect of laser window coating, which is not typically done in PLD due to technical challenges, which likely plays a role in the reproducibility considering the well-known stoichiometry modulation of STO with laser fluence.

Figure 2 shows characterizations of the selected films from Figure 1 to understand the surface quality, crystal structure, and chemistry of the $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ films used for machine learning. Figure 2a-c show the surface topography measured by atomic force microscopy (AFM) of Ti rich ($N = 7$), stoichiometric ($N = -1$), and Sr rich ($N = -5$) films. The films have a roughness of 0.547 nm, 0.492 nm, and 0.215 nm for $N = 7$, $N = -1$, and $N = -5$, respectively. This corroborates with the streaky RHEED patterns in Figure 1e-g. X-ray diffraction (XRD) θ - 2θ scans shown in Figure 2d confirm that the films are c-axis oriented $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ with no phase separation into the binary components TiO_2 or SrO due to sequential deposition from these targets. Figure 2e shows a detailed θ - 2θ scan around the STO (002) reflection. We find an increase in the out-of-plane lattice parameter that is typical of off-stoichiometry STO films³¹. Finally, XPS spectra of the Sr 3d and Ti 2p core-levels (Figure 2f-g, respectively) confirm the presence of only Sr^{2+} and Ti^{4+} expected for STO. The binding energy for each core-level linearly increases with decreasing x . The Sr $3d_{5/2}$ peak has binding energy of 132.8 eV, 133.1 eV, and 133.4 eV for $x = 0.69$, 0.50, and 0.38,

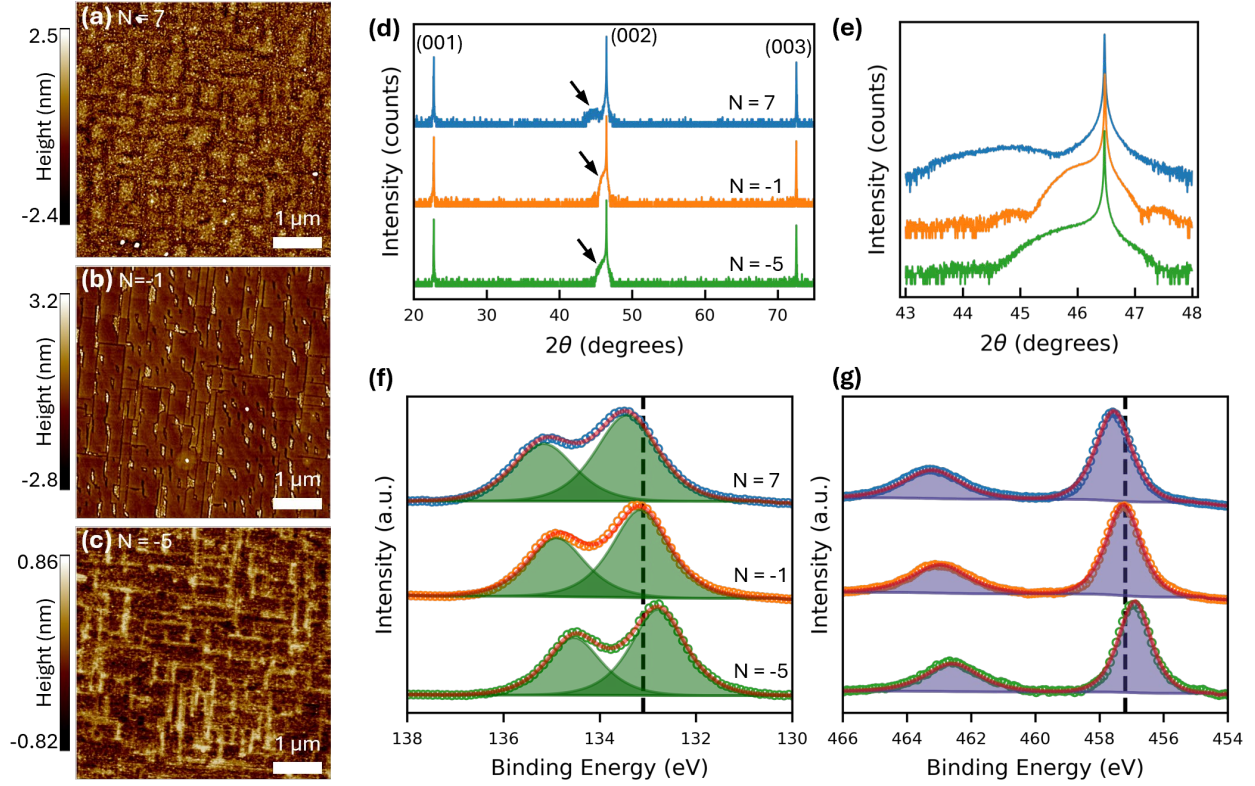


Figure 2. Characterization of selected $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ films that are Sr rich ($N = -5$), stoichiometric ($N = -1$), and Ti rich ($N = 7$). a-c) Film topography from atomic force microscopy images displays sub-nanometer roughness. d) Wide angle θ - 2θ XRD scans confirm (001) oriented $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ films with no impurity TiO_2 or SrO phases present. e) θ - 2θ XRD scan of the SrTiO_3 (002) reflection shows c-axis expansion the film when for Sr-rich and Ti-rich cases. XPS of the f) Sr 3d and g) Ti 2p core-levels show examples of the calculated peak areas used to measure the atomic fraction of Sr in each film. XPS also shows a linear shift in the binding energies of Sr 3d and Ti 2p with stoichiometry. The dashed lines mark the typical binding energies of 133.1 eV and 457.2 eV for the Sr 3d_{5/2} and Ti 2p_{3/2} peaks in SrTiO_3 , respectively.

respectively. Similarly, the Ti 2p_{3/2} peak has a binding energy of 456.9 eV, 457.2 eV, and 457.6 eV for the same x values, respectively. These shifts are most likely due to variation in charge compensation that is not fully corrected by aligning the C 1s peak to 285.0 eV and do not indicate changes in Sr or Ti ionic charge state. The shifts are also consistent with an increased inner potential for a Ti-rich surface and decreased inner potential for a Sr rich surface, which should affect the kinetic energy of the photoelectrons as they escape the surface³². However, the use of an electron flood gun to mitigate charging prevents us from drawing any clear conclusions in this

regard. Note here that the fit areas of these XPS peaks are the ones used for each sample to calculate the Sr atomic fraction used as the machine learning target. It is worth noting that the value of x extracted from XPS fitting is not necessarily a precise estimate of film composition due to surface effects, such as the preferential accumulation of Sr or Ti on the surface of Sr-rich or Ti-rich films, respectively³³. However, the relationship between absolute composition, such as what might be measured via Rutherford backscattering, and XPS-derived composition should be monotonic and does not affect the validity of the ML model.

Using the dataset we generated for $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ films, we trained a gated convolutional neural network (CNN) to extract features from the RHEED images to predict the Sr atomic fraction in the films. Instead of using standard convolutions, we chose gated convolutions^{34,35} to leverage the effective spatial attention mechanism created by the learned gate weights. Gated convolutions effectively “turn off” image pixels that are found to be unimportant to the model’s prediction, thereby providing additional information useful for interpreting which image features are important. Figure 3a shows the model architecture which consisted of 3 sequential blocks of gated convolutions (Gated-Conv) and max pooling layers (Pool) which is finally flattened and passed through a fully connected block (FC) which consists of 2 gated linear units (GLU) with dropout to output the Sr atomic fraction prediction. The gated convolution splits (Chunk) the output of a standard 2D convolution (Conv2d) along the channel dimension after batch normalization (BatchNorm), applies a nonlinear activation function to one half (ReLU), a sigmoid function to the other (the gate), and multiplies the results to rescale the convolution activations based on the learned spatial attention of the gate. The model hyperparameters were optimized through a

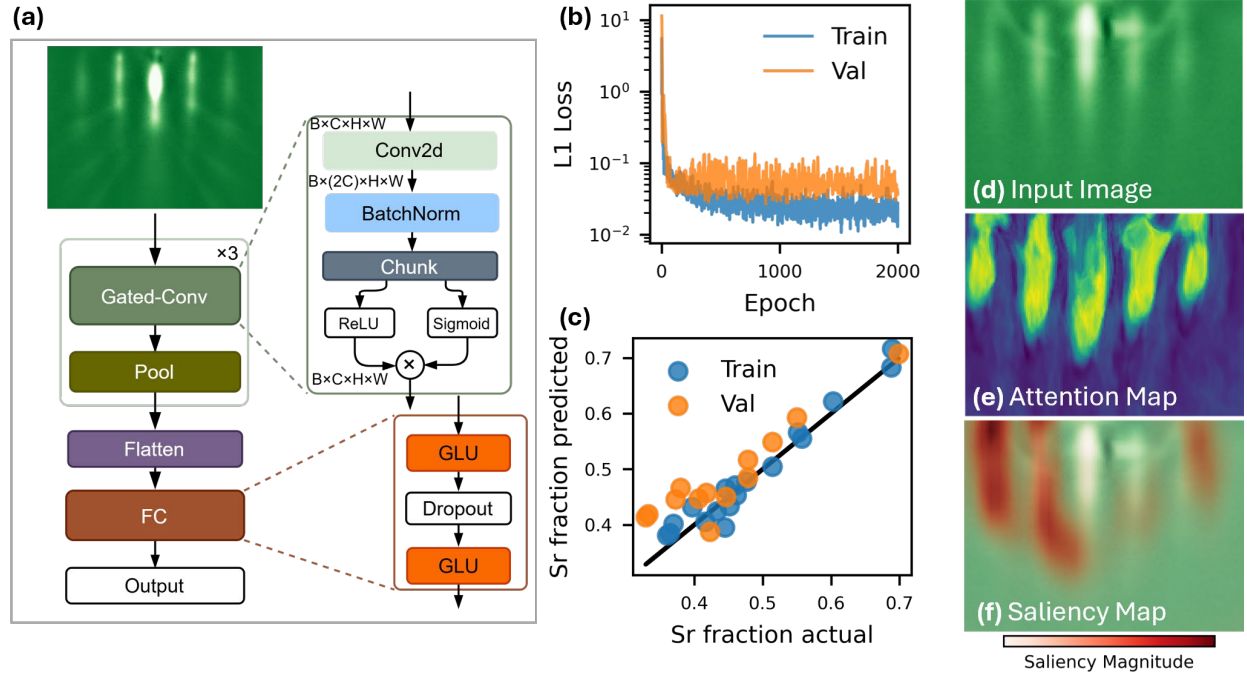


Figure 3. Deep learning of RHEED images for prediction of Sr atomic fraction in $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ films. a) The machine learning model extracts deep features by passing RHEED images through 3 sequential gated convolutional layers, where the gate in each layer acts as an attention mechanism that emphasizes the most important feature activations. b) Learning curve of L1 loss vs. training epoch shows convergence. c) Model predicted vs. actual Sr atomic fraction with coefficient of determination $r^2_{\text{train}} = 0.956$ and $r^2_{\text{val}} = 0.867$. For an example input image (shown in d), the e) attention map derived from the mean gate weights highlights the important regions in the image and the f) saliency map indicates the pixel importance of the $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ (01) and (02) RHEED streaks for predicting the Sr atomic fraction.

3-fold cross-validation tuning procedure, with both image and target augmentation, to identify the optimal configuration. Figure 3b shows the L1 loss curve of the optimized model, demonstrating its convergence. Figure 3c shows the model predicted vs actual Sr atomic fraction for the training and validation sets. The model achieves a coefficient of determination of $r^2_{\text{train}} = 0.956$ and $r^2_{\text{val}} = 0.867$, respectively, demonstrating reasonable accuracy on this small dataset of 31 samples. Thus, the model can effectively learn deep features from the RHEED images that are highly correlated to the film stoichiometry.

To understand what regions of the RHEED images the model correlates to the Sr atomic fraction, we visualize attention maps and employ gradient-weighted class activation mapping, Grad-CAM++³⁶ modified for regression, to visualize the pixel saliency. Using an example image (Figure 3d), we visualize the attention map by averaging all 128 gate channels of the final convolution layer, shown in Figure 3e. The attention mechanism primarily gives large weight to the diffraction streaks and small weight to the background in the lower portion of the image and between the streaks. Interestingly, the attention mechanism gives no weight to an image artifact created by a defect in the RHEED screen (seen as the dark spot in Figure 3d), which is consistent because it has no bearing on the film properties. Finally, we visualize the pixel saliency, which highlights the pixels that are most influential to the prediction. A large saliency magnitude indicates that the pixel value (intensity) at a given location greatly affects the prediction. Figure 3f shows the saliency map, with most of the saliency weighted towards the (01) and (02) diffraction planes, giving little weight to the central streak or spectral reflection. Generally, the model predictions are influenced by the diffraction streaks and the greatest salience typically involves the pixels containing or near the (01) and (02) diffraction planes.

Finally, we can use the information learned from the saliency maps to guide manual analysis of the RHEED images to search for empirical correlations to corroborate/explain the model

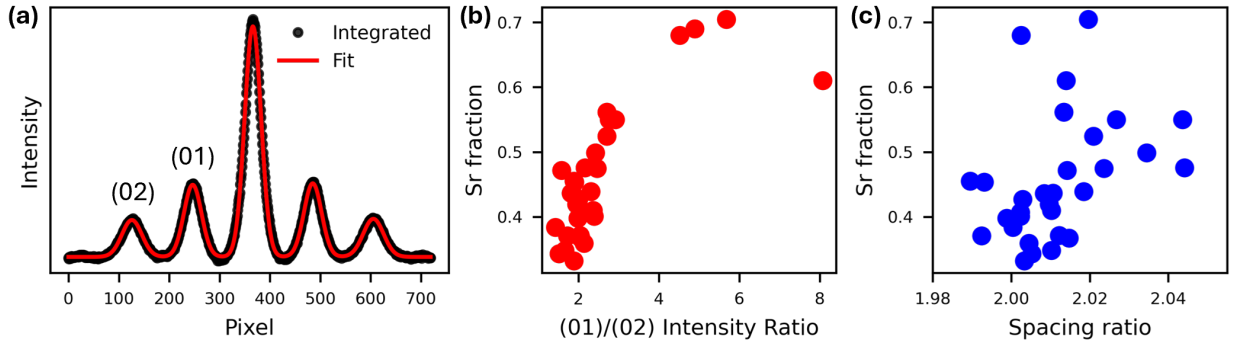


Figure 4. Empirical RHEED features suggested from salience analysis of machine learning model that are highly correlated to the atomic fraction of Sr in $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ thin films. a) Example of an integrated RHEED image for one sample ($N = 6$, $x = 0.419$) with a multiple Gaussian peak fit. b) The intensity ratio of the (01)/(02) RHEED diffraction streaks and the c) The ratio of the pixel spacing from the central diffraction plane of the (02) and (01) streaks. Both empirical metrics were discovered with the aid of the deep features learned from the model.

predictions. We integrated the RHEED images and fit Gaussians to the diffraction peaks to examine the relative intensity and spacing (i.e., inversely proportional to the in-plane lattice parameter) of the (01) and (02) planes. Figure 4a shows an example of the integrated RHEED with the multi-Gaussian fit. Indeed, we find empirical correlations between the (01) and (02) peak parameters and the Sr atomic fraction. Figure 4b shows the (01)/(02) intensity ratio vs Sr fraction. We clearly observe a strong positive correlation, although with an outlier point. Similarly, Figure 4c shows the ratio of the pixel spacing of each peak relative to the central diffraction spot, i.e. the ratio of the 1st and 2nd order reciprocal lattice vectors $(C_{(02)} - C_{(00)})/(C_{(01)} - C_{(00)})$ where C is the peak center. We also observe a positive correlation with this metric.

Comparing the intensity of the specular reflection and 1st order diffraction peaks during SrTiO_3 growth has been used to determine the SrO vs. TiO_2 termination^{37,38}. To our knowledge, these empirical correlations between diffraction intensity/spacing ratios of the (01)/(02) peaks and Sr/Ti ratio in STO have not been noted in the literature, likely due to a lack of data which spans such a broad range of stoichiometry. Here we synthesized films with a two-target synthesis approach to access a wide variety of stoichiometry for training but these empirical correlations, as well as the

trained model, should be applicable to STO films synthesized using a traditional single target approach.

We speculate that the cause of these variations is due to dynamical diffraction and distortions from the ideal SrTiO_3 lattice. Kikuchi lines (formed by multiple scattering) have been used to measure the oscillation of the inner potential during STO growth which is correlated with the surface termination, and that the incident angle is an important to understand the RHEED pattern³². Considering the present successful application of deep learning, we encourage experimentalists to routinely report the incident angle of the electrons, when possible, to assure comparable RHEED patterns between groups. Deep learning can currently be used internally within research groups but may struggle to predict on images between groups. To overcome this, a database of simulated RHEED patterns could be generated to train general ML models for RHEED – but such dynamical simulations are a challenging area of active research.

In this work, we demonstrated the efficacy of deep learning for regression of continuous valued properties of thin films using RHEED images acquired during synthesis. We synthesized 31 $\text{Sr}_{2x}\text{Ti}_{2(1-x)}\text{O}_3$ thin films using PLD, recorded RHEED images for each sample during deposition, and measured the atomic fraction of Sr using XPS to generate a dataset for analysis. Using this dataset, we trained a gated convolutional neural network that can successfully predict the Sr atomic fraction from the RHEED images, with an accuracy that is comparable to post-growth XPS measurements. Explainable AI techniques indicated a correlation between the (01) and (02) diffraction planes and the film stoichiometry. Guided by these machine-learned features, manual analysis revealed previously unknown empirical correlations between the relative intensity and spacing of these diffraction peaks. Our findings highlight the potential for deep learning with RHEED images as a “simple surrogate” measurement that can substitute for more in-depth *ex situ*

properties measurements that are affected by crystal structure. Such an approach can accelerate both traditional and autonomous synthesis campaigns by deriving greater value from otherwise qualitative routine diagnostics where correlations are not obvious or previously known.

CONFLICT OF INTEREST

The authors declare no competing interests.

AUTHOR CONTRIBUTIONS

S.B.H.: investigation (lead); data curation (lead); methodology (equal); software (lead); visualization (lead); writing—original draft (lead); writing—review & editing (lead). Conceptualization (supporting). **P.T.G.:** Conceptualization (equal); investigation (supporting); software (supporting); writing—original draft (supporting); writing—review & editing (supporting). **C.M.R.:** Resources (equal); Methodology (equal); writing—review & editing (supporting). **R.K.V.:** writing—review & editing (supporting); Conceptualization (equal); Investigation (supporting) **R.B.C.:** Conceptualization (equal); writing—review & editing (supporting), Resources (equal). All authors read and approved the final manuscript.

DATA AVAILABILITY

The data that support the findings of this study are openly available at <https://github.com/sumner-harris/Deep-Learning-with-RHEED.git>

CODE AVAILABILITY

The code that supports the findings of this study are openly available at <https://github.com/sumner-harris/Deep-Learning-with-RHEED.git>

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Supporting Information Available

Experimental methods; Machine learning model training and data preprocessing.

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TOC Graphic

